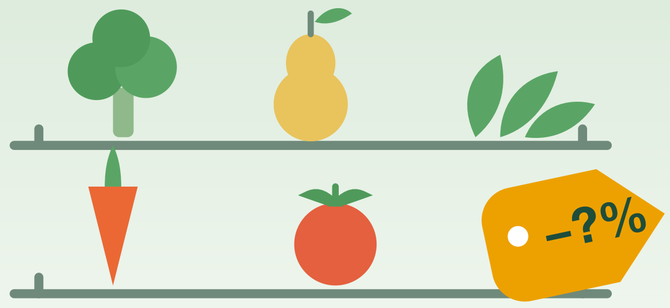
HKU MAIB 7002 · GROUP Q · MACHINE LEARNING FOR PRICING

# Cut the price, or trust the shelf?

We taught a model to predict what a perishable will sell tomorrow at any discount. Then we discovered it would discount everything, and had to teach it to read the shelf first.

**Aashish Omprakash Pareek · Yuanyuan Yin · Lingyu Chen** · Dingdong FreshRetailNet-50K, a public dataset of 50,000 store-product histories




6 pm · the shelf is still half full · what is tomorrow's price?

## 01 A decision every evening, for every product

### 6 pm, the shelf is half full

Tomorrow's discount must be set tonight, before anyone knows how much will sell.

### Two ways to lose

Hold the price and stock may be thrown away. Cut it and margin goes on sales that would have happened anyway.

### Our question

For one product, in one store, on one day: how much sells tomorrow at each discount, and should we discount at all?

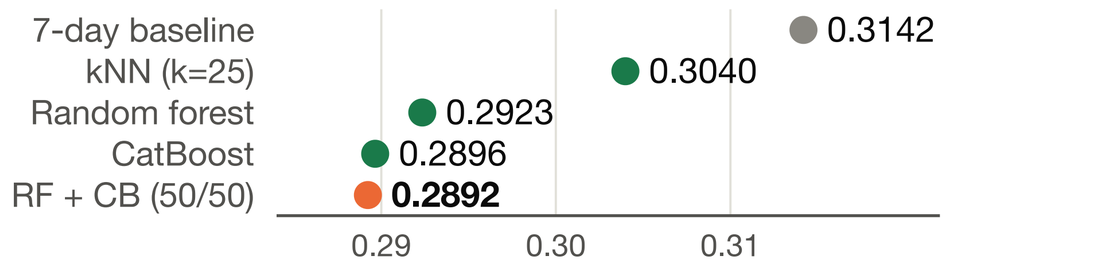
## 02 The raw material

**312** product histories in five stores, picked because their discounts vary

**28,080** store-product-days; one unseen week of 2,184 kept for the test

**45%** of days have an empty-shelf hour, a clue we later put to work

## 03 Forecasting results




Test MAE over 2,184 forecasts: 312 products × 7 held-out days (lower is b

**7.9%** lower MAE than the 7-day average with the 50/50 random forest + CatBoost blend; squared error 22% lower

**What we learned:** every model beat the no-ML average, the three tree models tie within noise, and choosing the inputs mattered more than any tuning dial. Dropping the weather was the single biggest gain.

## 04 Model choices

### kNN · 25 neighbours

Averages the 25 most similar past days; dropping weather cut its error from 0.327 to 0.312.

### CatBoost · 500 trees

Depth 5, learning rate 0.03: small and slow on purpose, because boosting over-fits.

### Random forest · 300 trees

Depth 10, 3 rows per leaf; all 16 settings beat the baseline every week.

### RF + CatBoost · 50/50 — FINAL MODEL

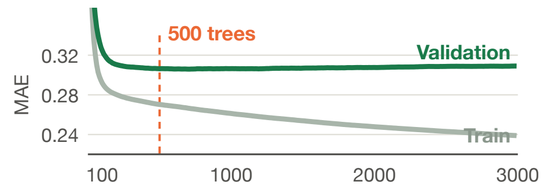
Averages both forecasts; errors correlate at 0.97, so the gain is small but consistent.

**Random forest: validation MAE across 16 settings**

| max depth \ min rows per leaf | 3 | 5 | 10 | 20 |
|---|---|---|---|---|
| 10 | 0.3024 | 0.3027 | 0.3035 | 0.3052 |
| 15 | 0.3030 | 0.3031 | 0.3032 | 0.3049 |
| 20 | 0.3043 | 0.3036 | 0.3034 | 0.3048 |
| None | 0.3044 | 0.3034 | 0.3031 | 0.3051 |

rows: max depth · columns: min rows per leaf · best outlined

**CatBoost: error as trees are added**




## 05 The twist: a good forecaster is a reckless pricer

**Ask the model "what if I discount?"** and it answers: deeper is always better. At a 0.6 rate it predicts 2.3× the sales.




It learned that from clearance days, which are fewer than 1% of the data. Left alone it would discount on 98% of days.

**So we put the shelf before the model.** Three stock rules decide first; the model only chooses among discounts the product has actually run.

### Sold out today?

Then no discount tomorrow: the shelf is clear.

### A week with no sell-out?

Then a discount is required; the model picks how deep.

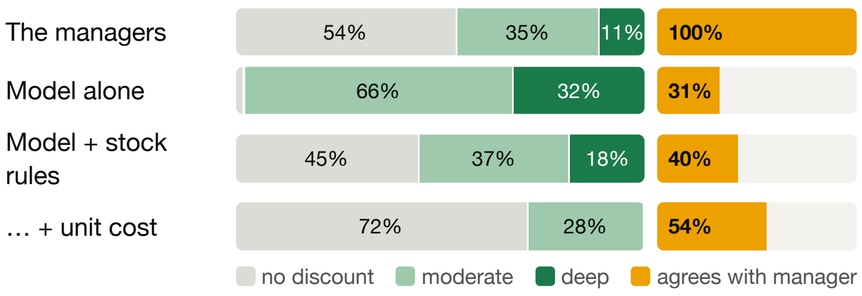
### Otherwise ask the model

Only about discounts this product has actually run.

### Choose

Highest predicted value, net of unit cost, if the gain is worth it.

**What each policy recommends** on the test week, and how often it lands in the same band as the real managers.




Stock rules decide 53% of days before the model is asked. Agreement rises from 31% to 40%, and to 54% once a unit cost makes deep discounts cost something.

## 06 The challenges we found most interesting

### The shelf lies

On 45% of days the shelf was empty for part of the day, so recorded sales are lower than real demand. The model learns from censored numbers.

### Sales without units or prices

The data is normalised: 1.2 is not 1.2 kilograms or dollars. "Value" is sales × discount rate, a proxy, so profit cannot be computed.

### Managers are not random

Discounts were chosen on days managers expected demand. Part of the lift the model sees is their judgement, not the price cut. Correlation, not cause.

### No counterfactual

We only ever see what sold at the price the manager chose. Whether our policy sells more or wastes less needs a real experiment.

### Weather made things worse

**0.327 → 0.312**

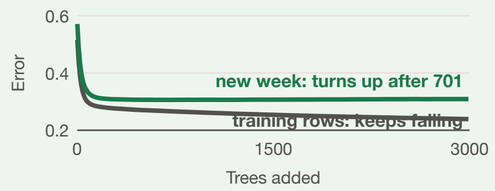
Removing rain, temperature and humidity cut the error more than any amount of tuning did.

### Deep discounts are the hard days




Error by tomorrow's discount: every model under-predicts the spike on exactly the days a pricing tool cares about.

### Boosting over-fits in plain sight




Training error keeps falling; error on a new week turns up after 701 trees.

### One city, one week, one guess at shelf life

Five stores in one city and a single test week that has been scored for four models. The "no sell-out for a week means old stock" rule assumes a shelf life the data never states.

Data: Dingdong-Inc/FreshRetailNet-50K on Hugging Face, CC BY 4.0 · scikit-learn and CatBoost · Errors are mean absolute error in normalised sales units

Full notebook, code and Docker image: github.com/Lisayinyy/MAIB_7002, branch pricing-ml-ap